I am fascinated by the duality between the richness of language and the computational mechanisms behind it. Classically, linguistics and formal language theory explore this connection in elegant terms. The computational properties imbued in grammars allow various types of linguistic structure to emerge. Languages can be understood as the observable outcomes of latent computational systems. However, when it comes to the recent success of neural networks, this perspective is often lost. Even when systems perform well, their computational abilities and representations are often not well understood on a conceptual level. Closing this gap is a fascinating goal for both practical and foundational reasons. I believe such understanding paves the way for principled improvement and interpretability of NLP models. It also has the potential to refine and expand theories of the structure of formal and natural languages. I view the following research questions as constituting different and interesting perspectives on this underlying goal.

1) What is the role of latent structure in building NLP models that generalize?
During my junior year, I became excited about the idea of trying to improve the **compositional generalization** of network networks by baking in mechanisms for **latent linguistic structure**. In one case, this meant hacking together a **differentiable pushdown automaton** (Hao et al., 2018) to solve tasks like string reversal (conventional RNNs fail on this). With natural language, we showed that stack models could be leveraged to improve agreement prediction on syntactically complex natural language and induce unsupervised parsed trees with nontrivial resemblance to gold-standard semantic trees (Merrill et al., 2019). In another project, we built a competitive neural parser around **Tree Adjoining Grammar** (TAG; Kasai et al., 2018), which improved the utility of the parses for downstream semantic tasks compared to baselines with other formalisms. The mild context-sensitivity of TAG allowed our parser to transfer from declaratives to questions with only a few examples. Both projects aimed to use the rich theory of formal languages and grammars to improve the structural representations within neural networks. However, there are challenges to making these approaches practical: sophisticated formalisms make data collection more expensive (in the supervised case), and it is not clear that the model will learn to use memory as the designer envisions it will. Moving forward, I am curious about advancements in methods for modeling latent structure, and how the priorities we assign to generalization abilities at evaluation impact our assessment of these approaches.

2) How can we understand the representations within neural networks on their own terms?
A related problem to (1) is exploring the representations inside standard NLP models, rather than developing extensions of them. On one hand, there is conventional wisdom in linguistics that hierarchical inductive bias is necessary for modeling language. On the other hand, neural network models (with no obvious such bias) seem to perform well on many NLP tasks. Resolving this mysterious disconnect has the potential to challenge conventional thinking on several fronts: perhaps RNNs are more powerful than we thought, modeling language doesn't require as much structure as assumed, or our evaluations of NLP models are broken (Given current results, a case can probably be made for all of these to some degree). With this question in mind, I developed a theory of **formal language capacity** for neural sequence models (Merrill, 2019) by analyzing the capacity of networks where all activation functions are **saturated** (pushed to their extreme values). Under this assumption, I derived differences in capacity for different architectures. Interestingly, these results predict empirical differences in learnability for trained networks. For example, GRUs reduce to finite-state automata, whereas the LSTM becomes a machine resembling a counter automaton. Experimentally, LSTMs can learn to stably model counter languages like $a^n b^n$ and 1-Dyck, whereas GRUs cannot. This led to the hypothesis that saturated networks describe some inherently learnable class of grammars for a neural network architecture.

This hypothesis motivated a natural follow-up question. Rather than keeping the connection between saturated networks and learnability as an empirical observation, I wanted to see if there was some theoretical explanation rooted in the **dynamics of training**. In Merrill et al. (2020), we show that properties of transformers (and related networks) lead the parameter norm to continually grow over training, which in turn causes the network to increasingly approximate a saturated network over time. Empirically, we analyzed data from the training of T5, a large transformer model, finding that the norm increases over the course of training, and that this corresponds to significant saturation of the representations within the pretrained network. A further goal of mine is modifying architectures or training algorithms to amplify inductive bias towards saturation. I would like to test the effect of increased saturation on model performance, and also explore ways to use the emergent discrete structure (such as hard attention maps) to prune unnecessary weights or visualize model decisions.

A related topic I have studied is the relation of rational RNNs (RNNs structured as weighted finite automata) to LSTMs. I proved that saturated LSTMs are fundamentally stronger than rational RNNs (Merrill et al., 2020), although in certain cases the weakness is alleviated by stacking RNN layers. We developed theoretical tools for these types of formal language questions by reducing them to questions about the rank of Hankel matrices. Practically, rational RNNs address the goal of leveraging continuous optimization to infer finite-state classifiers from data. I am broadly interested in this question and its connections to efficiency and interpretability.

3) Is it possible to acquire propositional semantics from unsupervised language modeling?
This direction relates more to the **philosophy of AI** rather than deep learning itself. In a recent project, I aimed to recast this question about NLP pretraining in formal terms, drawing on foundations in logic and theoretical computer science. Defining understanding as emulating representations that preserve latent semantic relations between phrases, I proved that distributional cues called *assertion queries* are sufficient for understanding text if its semantics is sufficiently simple (i.e. referentially transparent). However, I prove that constructions like variable binding make this impossible in the sense that emulation becomes **Turing-uncomputable**. This analysis restates an open question about the limits of current NLP pretraining in the language of computability, and reaches different conclusions based on the complexity of the underlying language. I interpret these findings as an impossibility result that, in general, semantics cannot be emulated from ungrounded text alone. Moving this analysis from formal to natural languages seems to only make the task harder, introducing intensionality, ambiguity, and other challenging features of language. In the future, I hope to examine how this impossibility result bears on finer-grained questions like the complexity of emulating semantics over a bounded support of text. More generally, I am interested in efforts to extend paradigms from formal semantics to theorize or evaluate language understanding in NLP systems.

4) What do neural network language models encode about linguistic change?
Language can be understood not just as a static computational system, but one that changes over time as it is transmitted to new speakers. I am interested in to what degree deep learning systems are sensitive to this diachronic linguistic change. In the past, I have explored how LSTM models represent syntactic change (Merrill et al., 2019). One basic question I want to look into is how GPT-3 can generalize text between different historical periods of a language (for example, rewriting a famous piece of modern English in Old English).

Research by Tal Linzen and Sam Bowman to explore the linguistic capabilities of NLP models aligns with many of these questions. I would be interested in being co-advised by Tal Linzen or Sam Bowman along with a theoretician like Mehryar Mohri or Chris Barker to continue my work at the intersection of NLP and computational and linguistic theory. I would also hope to collaborate with researchers in deep learning theory.